

EX 6

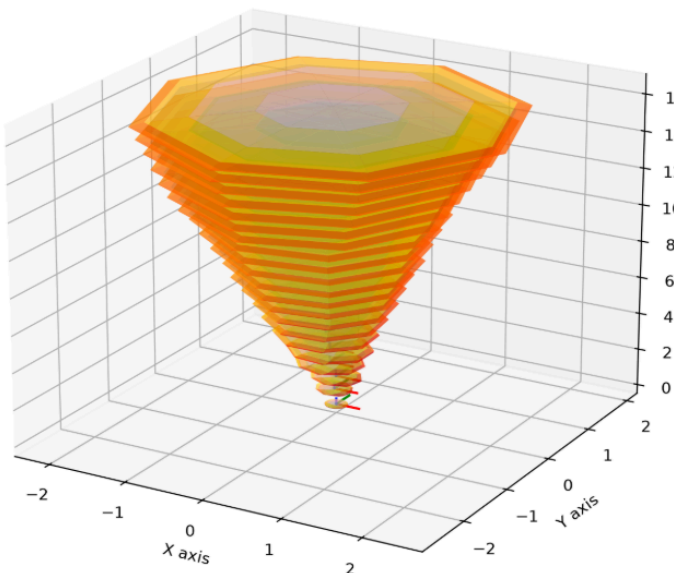
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[Github](#)

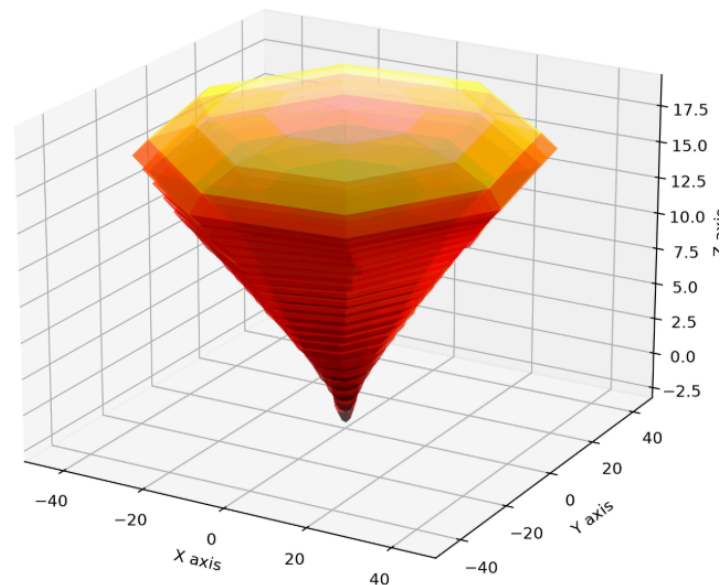
6.1:

As we can see, covariance of the bundle are very sensitive to the prior noise:

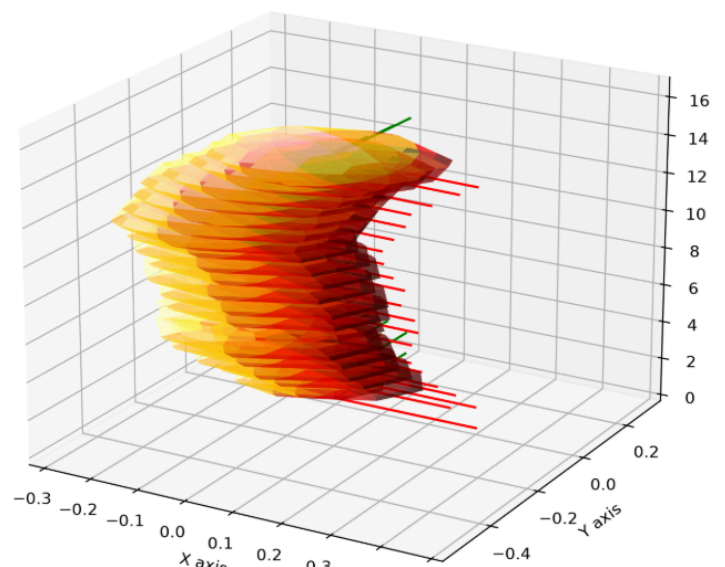
6.1 - Bundle 0→20 | prior noise: $I \times 0.05$ ($\sigma=0.05$)
Frame locations with marginal covariances



6.1 - Bundle 0→20 | prior noise: Unit matrix ($\sigma=1.0$)
Frame locations with marginal covariances



6.1 - Bundle 0→20 | prior noise: $I \times 1e-6$ ($\sigma=1e-6$)
Frame locations with marginal covariances



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--- Relative pose & covariance for c0=0 → c_k=20 ---
Graph size before optimization: 1022
Initial estimate size: 302
Window 0->20: factors=1022, error before=1554.69, after=218.37

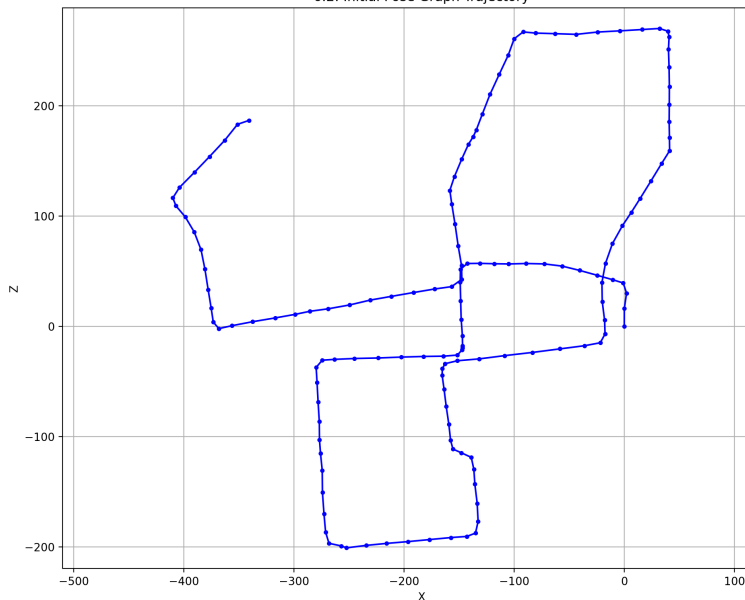
Relative Pose between keyframes c0 and c20:
R: [
    0.996652, 0.0396538, 0.0714984;
    -0.0395722, 0.999213, -0.00255786;
    -0.0715436, -0.000280057, 0.997437
]
t: 0.123413 -0.405426 16.1642

Conditional covariance P(c20 | c0):
[[ 0.00000321 -0.0000006 0.00000645 0.00001094 0.00010361 -0.00002245]
 [-0.0000006 0.00000339 -0.00000388 -0.00011733 -0.00003545 0.00017431]
 [ 0.00000645 -0.00000388 0.00003306 0.00007382 0.00025262 -0.00010556]
 [ 0.00001094 -0.00011733 0.00007382 0.00470863 0.00091268 -0.00769985]
 [ 0.00010361 -0.00003545 0.00025262 0.00091268 0.00412112 -0.00240668]
 [-0.00002245 0.00017431 -0.00010556 -0.00769985 -0.00240668 0.02690397]]

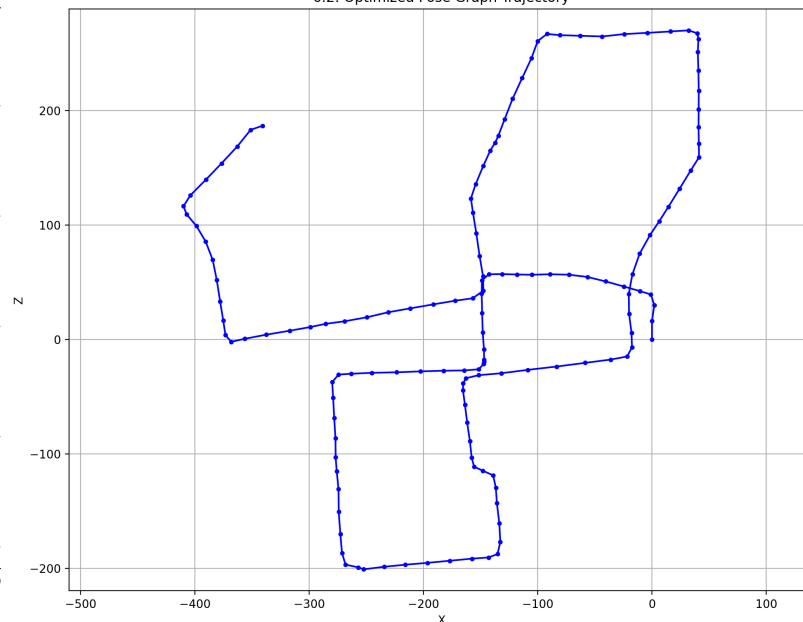
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6.2:

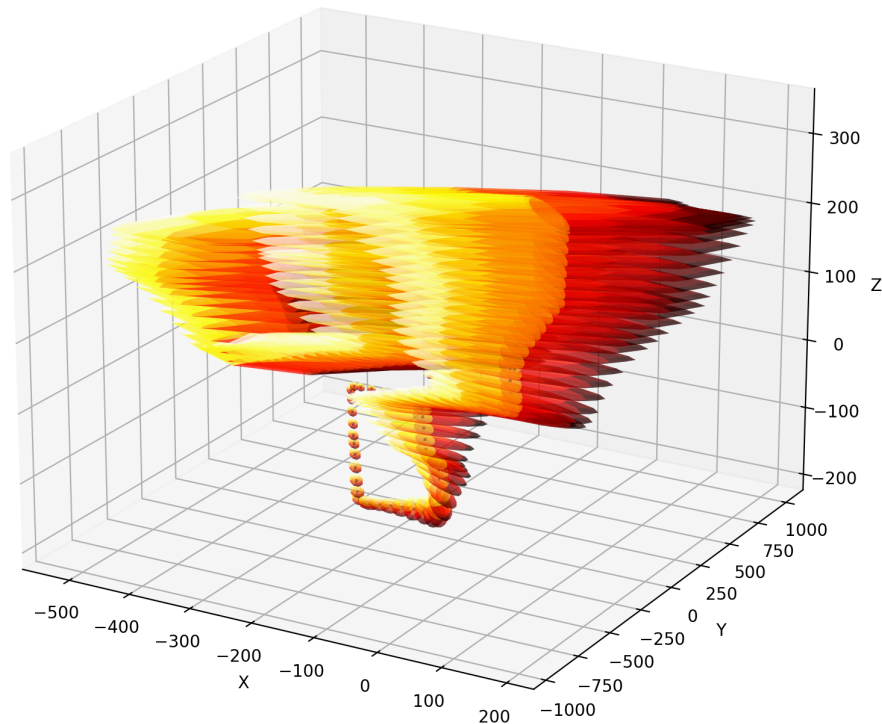
6.2: Initial Pose Graph Trajectory



6.2: Optimized Pose Graph Trajectory



6.2: Optimized Pose Graph With Marginal Covariances



A pose graph was constructed using the selected keyframes as nodes and the relative motions estimated in the previous section as Between-factor constraints. The covariance matrix computed for each bundle window was used as the uncertainty model of the corresponding constraint. A natural initialization was obtained by chaining the relative transformations between consecutive keyframes, starting from the first keyframe fixed at the origin.

The resulting initial trajectory is shown in **initial_pose_graph**. The graph contained 143 pose variables and 143 factors, corresponding to the largest connected component of the estimated trajectory (142 relative constraints).

After optimization, the trajectory shown in **optimized_pose_graph** remained visually identical to the initial trajectory. This is expected because the initial estimate was generated directly from the same

relative pose measurements used as graph constraints, causing the factor residuals to be nearly zero already before optimization. Accordingly, the factor graph error before optimization was 0.0 and remained 0.0 after optimization.

The final marginal covariance estimates are shown in **pose_graph_covariances**.

As expected, the uncertainty generally increases with distance from the anchored first keyframe due to the accumulation of uncertainty along the chain of relative motion constraints and the absence of loop-closure measurements that could reduce the drift.

The covariance visualization appears large because the marginal covariance is computed over the full six-dimensional pose state (translation and rotation). The increasing spread reflects the accumulation of uncertainty along the pose chain in the absence of loop-closure constraints.